

9(a)	any four from: <i>motor / neurone</i> 1 resting potential higher ; 2 -70 mV (motor) versus -90 mV (muscle) ; 3 smaller change in membrane potential / smaller depolarisation ; 4 110 mV (motor) versus 130 mV (muscle) ; 5 action potential / depolarisation, takes less time / is faster ; 6 (action potential) 2.7 ms (motor) versus 4 ms (muscle) or (depolarisation) 1 ms (motor) versus 2 ms (muscle) ; 7 hyperpolarisation / <u>relative</u> refractory period, occurs ;	4
9(b)(i)	any three from: 1 transverse / T- , tubules depolarised ; 2 Ca^{2+} <u>diffuse</u> out of sarcoplasmic reticulum ; 3 Ca^{2+} bind to troponin ; 4 tropomyosin, moves from / exposes, binding sites ; 5 for myosin on actin ; 6 cross-bridges form / myosin (head) binds to actin ;	3

Question	Answer	Marks
9(b)(ii)	any one from: 1 cross bridges break / myosin heads detach, at different times ; 2 AVP ;	1

Question	Answer	Marks
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